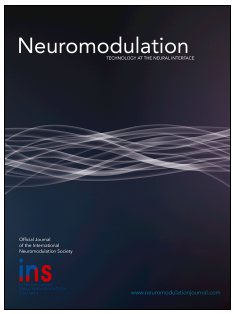
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Acute Effects of Transcutaneous Spinal Cord Stimulation (tSCS) on Gait Symmetry in People with Chronic Stroke

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Title: Acute Effects of Transcutaneous Spinal Cord Stimulation (tSCS) on Gait Symmetry in People with Chronic Stroke

Running title (40 characters or less): Acute tSCS Effects on Post-Stroke Gait

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Abstract

Objectives

To determine whether transcutaneous spinal cord stimulation (tSCS) modulates gait symmetry after a single bout of 25 minutes of treadmill gait training and to establish whether a particular combination of tSCS frequency and intensity elicits greater gait symmetry changes.

Materials and Methods

In each of six sessions, 20 chronic stroke participants received 25 minutes of treadmill gait training while they received either SHAM stimulation or one of four unique STIM (tSCS) parameter combinations: 30 or 50Hz frequency, and intensity at participant tolerance (TOL) or 30% of minimum resting motor threshold (RMT_{min}). In each session, our primary outcomes were step length (m) and swing duration (sec) symmetry during the 10-meter Walk Test (10MWT) PRE and POST treadmill training. We also computed an "overall" symmetry measure, the Combined Gait Asymmetry Metric (CGAM), and 10MWT gait speed. PRE to POST-intervention changes and how they differ among stimulation parameters were assessed via the *Timepoint x Intervention* interaction effect of a two-way repeated measures ANOVA. A complementary exploratory analysis was performed to determine individualized "best" vs "worst" stimulation parameters compared to SHAM.

Results

Among the five interventions tested, we found no statistically significant difference in the PRE to POST change in the primary outcome variables – step length symmetry (*Timepoint x Intervention* effect, p=0.0564) and swing duration symmetry (p=0.6215) – or in CGAM (p=0.5350) or 10MWT gait speed (p=0.4828).

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Conclusions

We found that 25 minutes of tSCS plus treadmill training did not significantly change gait symmetry or speed compared to treadmill training alone. These findings also suggest that in a single session setting there may not be a best frequency or intensity that is optimal for everyone. Future work should investigate whether person-specific stimulation parameters may yield single-day gait symmetry changes, as well as potential longitudinal effects of tSCS plus gait training.

Key words

Neuromodulation, spinal cord stimulation, transcutaneous, gait, symmetry, stroke

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Introduction

Current best clinical practice for recovery of gait function after stroke is moderate to high intensity gait training, often with a focus on reducing the asymmetry typically observed [1]. Moderate to high intensity gait training leads to a cascade of neurophysiological changes. Rat models have shown that bouts of similar intensity exercise induces neurogenesis and the release of growth factors such as brain-derived neurotrophic factor [2]. In humans, moderate to high intensity gait training has also yielded functional changes such as improved gait speed [3] and endurance [4–6].

However, patients with stroke frequently continue to demonstrate asymmetric gait patterns during participation in moderate to high intensity gait training alone. Over time, ambulation with these asymmetrical gait patterns can lead to pain, orthopedic dysfunction, inactivity, and reduced engagement in the community. Therefore, there is a need for multimodal rehabilitation strategy that may involve such interventions as drugs, exoskeletons, and/or stimulation to pair with gait training in a way that facilitates neuroplasticity and promotes restoration of more typical & efficient gait patterns, including symmetric gait.

One promising approach is transcutaneous spinal cord stimulation (tSCS). tSCS is a non-invasive neuromodulation technique that delivers electrical current via an adhesive electrode placed on the skin overlying the spinal cord. tSCS is thought to activate sensory afferent fibers within the dorsal roots of the spinal cord, producing trans-synaptic activation of spinal circuits that can facilitate motor output [7]. This mechanism of action is thought to be capable of facilitating recovery in a diverse set of neurologically impaired populations, including cerebral

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palsy [8], stroke [9], and spinal cord injury (SCI) [10]. Note that the study of tSCS for motor recovery has studied people with SCI most extensively [7].

However, there continues to be a paucity of evidence surrounding the most appropriate stimulation parameters that should be used to enhance gait training [11–13]. Additionally, in contrast to people with SCI, people with stroke often have intact sensation, thus each person tolerates a different stimulation intensity. Therefore, the limited findings regarding tSCS parameter selection in spinal cord injury may not directly translate to people with stroke.

Prior work by our group in people with stroke has examined how resting motor thresholds (RMT) vary with biphasic or monophasic waveform, the presence or absence of a carrier frequency [14], and person-specific characteristics, particularly body weight [15]. RMT is the person-specific minimum current intensity required to elicit a muscle response. tSCS dosing as a percentage of RMT provides a physiologically meaningful reference, typically stimulating below this threshold to avoid directly contracting muscles. The optimal intensity relative to the RMT to maximize functional benefit remains unknown, as our prior work did not investigate the effects of varying tSCS parameters on gait performance. The effect of varying another important tSCS parameter, burst frequency, on gait performance in persons with stroke also has yet to be fully investigated. Initial findings suggest that different frequencies may provide different benefits, such as targeting spasticity reduction at 50Hz [16] or gait characteristics directly at 30Hz [17,18].

Finally, the acute therapeutic effects of tSCS on gait in persons with stroke are relatively unknown. While recent preliminary evidence demonstrates effectiveness of tSCS to improve gait function when paired with longitudinal treadmill training [9,19], and in acute orthotic settings (i.e. while tSCS is actively applied) [21,22] , the potential benefits of a single bout of tSCS plus gait training are underexplored.

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Purpose & Hypotheses

There were two primary purposes of this study. The first was to determine whether tSCS plus treadmill training leads to superior improvements in gait symmetry (step length symmetry, swing duration symmetry, and Combined Gait Asymmetry Metric) and gait speed compared to treadmill training alone in a single session. Second, to determine whether a particular tSCS frequency and intensity combination leads to greater improvements in gait symmetry and speed than any other combination. We hypothesize that gait symmetry and speed will improve after a single session, and that there will be one combination of tSCS frequency and intensity that will yield greater improvements than the others.

Materials and Methods

ID	Sex	BMI	Type of Stroke	Chronicity (Years)	Paretic Side	Community Assistive Device	Community Orthotic Type
01	F	37.1	Hemorrhagic	2.2	L	SBQC	None
02	M	24.8	Ischemic	6.7	R	Forearm crutch	Articulated AFO*
03	M	36.3	Ischemic	9.5	R	SC for community, None household	None
04	M	26.8	Ischemic	2.7	R	None	None
05	F	41.2	Ischemic	4.3	R	SC	Carbon fiber AFO*
06	F	41.6	Ischemic	4.5	L	SC	None
07	M	35.5	Ischemic	3.8	L	SC	Solid AFO
08	M	34.4	Ischemic	2.0	L	SC	Carbon fiber AFO*
09	M	25.1	Hemorrhagic	9.4	R	LBQC	Solid AFO*
10	M	23.6	Ischemic	4.4	R	None	None
11	M	35.3	Hemorrhagic	4.4	L	SC	None
12	M	22.8	Hemorrhagic	2.7	L	SC	None
13	M	29.9	Ischemic	5.8	L	LBQC	Articulated AFO*

14	M	30.0	Ischemic	6.5	L	LBQC	Solid AFO*
15	F	31.4	Ischemic	5.7	R	None	Carbon fiber AFO*
16	M	23.9	Ischemic	4.7	R	None	Shoe Up
17	F	30.1	Hemorrhagic	3.28	L	SC	Carbon fiber AFO*
18	M	27.7	Hemorrhagic	5.5	R	SC	Articulated AFO in community* (none at home)
19	F	36.7	Ischemic	0.7	R	None	None
20	F	30.7	Ischemic	8.05	L	None	Carbon fiber AFO*

**Orthotic also used during the 10MWT. Note that no one used an assistive device during the 10MWT in this study.*

Table 1. Participant demographics. SC: straight cane, LBQC: large base quad cane, SBQC: small base quad cane, AFO: ankle-foot orthotic.

Participants

Twenty stroke survivors completed this study (Sex distribution: 13 m; Mean age: 53.3 ± 10.8 yrs; Mean weight: 90 ± 19.7 kg; Mean height: 1.70 ± 0.15 m) after providing informed consent in accordance with the Institutional Review Board of Northwestern University and the Declaration of Helsinki (STU00215009) and pre-registered at ClinicalTrials.gov (NCT05167786). Participants were included if they met the following criteria: ischemic or hemorrhagic stroke occurring at least 6 months prior to enrollment, age 18 years or older, functional ambulation category of two or greater, not currently receiving regular physical therapy, and physician approval. Participants were excluded if they met any of the following criteria: ataxia, history of multiple strokes, botulinum toxin injection in the lower extremity within the last four months, Modified Ashworth score of three or greater in the lower extremity, pregnant or nursing, using a powered, implanted cardiac device such as a pacemaker,

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defibrillator, or similar device, active cancer or cancer remission less than five years, pain due to active injury or infection, peripheral neuropathy, or neurological conditions that would impact the study.

Participants were excluded after a secondary screening session if they did not exhibit a spinal motor evoked response (sMER) in response to single pulse stimulation or were unable to tolerate continuous stimulation during walking of at least 25% of their lowest resting motor threshold (RMT_{min}). Three participants were eliminated from the study at this stage, leaving a cohort of 20 participants. Participant demographics are provided in Table 1.

Resting Motor Threshold (RMT) Assessment

Resting motor threshold (RMT) was quantified by surface electromyography (EMG) sensors (Delsys, Natick MA USA) at the start of each session as the lowest stimulation intensity at which a sMER was elicited. EMG responses were collected at 2,000Hz from the muscle bellies of five muscles bilaterally of each participant – the left and right rectus femoris (RF), vastus lateralis (VL), medial hamstring (HAM), tibialis anterior (TA), and medial gastrocnemius (MG). Starting at 10mA, each participant received five individual unmodulated biphasic 2ms pulses (1ms per phase) from the Digitimer DS8R constant current stimulator device (Welwyn Garden City, UK), spaced five seconds apart [14]. These five pulses were repeated, increasing stimulation intensity by 10mA, until either participant tolerance or 250mA (the maximum available current in most commercially available tSCS devices) was reached. The stimulation was applied via a 2 x 3.5” (~5.1 x 8.9 cm) rectangular electrode placed on the skin, with the top of the electrode aligned with the T11-T12 intervertebral space, resulting in the bottom of the electrode at or near the L1-L2 intervertebral space. Two ground rectangular electrodes were

placed bilaterally on the iliac crests. A splitter cable was used to connect the two ground electrodes to one output from the stimulator.

The RMT for each muscle was defined as the lowest stimulation intensity where three of the five pulses yielded a peak-to-peak amplitude EMG response ≥ 0.05 mV [14,22].

Session Design & Tested Stimulation Parameters

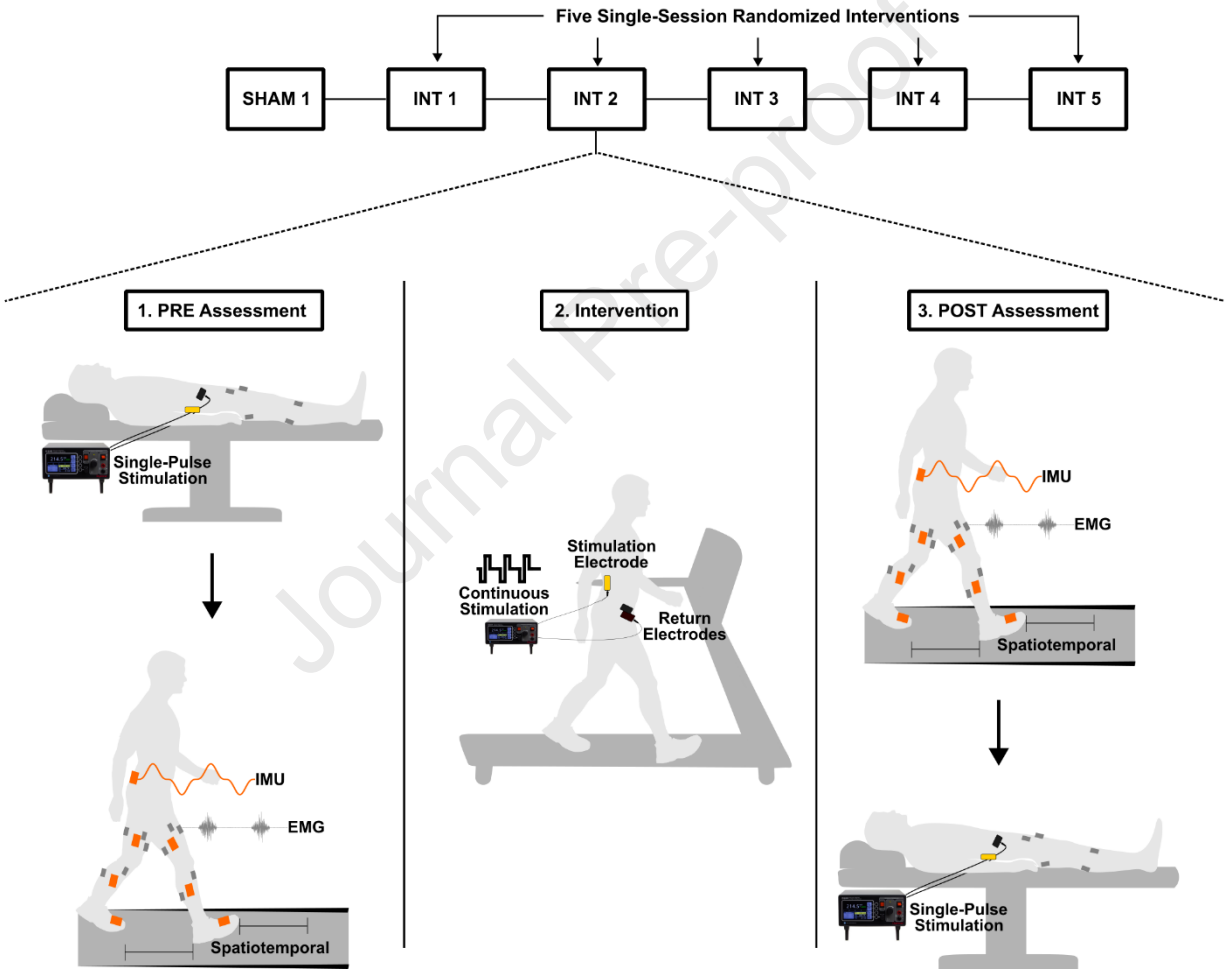


Figure 1. The study design. The first session, SHAM1, was dropped from analysis as a wash-in. In each of the remaining five sessions, the stimulation parameter was randomly assigned. In all sessions, participants first underwent a PRE assessment to

quantify RMT's, then a 10MWT. Next, they walked on the treadmill while receiving that session's intervention. Finally, they underwent a POST assessment, completing first the 10MWT, and finishing with RMT testing.

After the screening sessions, each participant completed six more sessions (Figure 1), each separated by at least 48 hours, generally completed over 2-3 weeks. In four of these sessions, they were randomized to receive tSCS at a different combination of intensity and frequency while walking on a treadmill with a training target of 40%-60% of heart rate reserve, monitored continuously throughout the session via heart rate monitor (Polar Electro, Kempele, Finland). To standardize tSCS dosage across participants, tSCS intensity was set to either 30% of the participant's lowest (minimum) RMT (RMT_{min}), or to their tolerance during that session (TOL), up to a maximum of 90% of RMT_{min} to avoid unwanted muscle contractions. 30% of RMT_{min} was used as in our prior work this is the highest intensity tolerated by all participants with a biphasic waveform [14]. tSCS frequency was set to either 30 Hz or 50 Hz, as they are the most common frequencies used in the literature [11,23]. Thus, one of four possible combinations of stimulation parameters was administered during each of the four treadmill training plus stimulation (STIM) sessions (30TOL, 50TOL, 30RMT_{min}, 50RMT_{min}). The actual stimulation intensities delivered in practice were: 30TOL: 54 ± 21% RMT_{min}, 50TOL: 52 ± 22% RMT_{min}, 30RMT_{min}: 27 ± 6% RMT_{min}, 50RMT_{min}: 27 ± 5% RMT_{min}. These %RMT values were obtained by visual inspection of the sMER. Note that visually determined RMT and RMT's determined via quantitative analysis differed. Dosages administered via quantitatively computed RMT were higher due to lower RMT_{min}: 30TOL: 63 ± 29% RMT_{min}, 50TOL: 55 ± 23% RMT_{min}, 30RMT_{min}: 28 ± 7% RMT_{min}, 50RMT_{min}: 31 ± 13% RMT_{min} (Supplemental Document 1). Potential reasons for this discrepancy include the subjectivity involved in the visual method compared to the quantitative RMT_{min}-based processing, and error inflation caused by discrete 10mA steps in stimulation intensity. The other two sessions followed the same protocol, except participants

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186 received SHAM tSCS (0mA stimulation) while on the treadmill. While assessors and therapists
187 were not blinded to the day's intervention, we attempted to blind participants as to whether each
188 session was SHAM or STIM, instructing them that "stimulation may feel different each day".
189 Participant awareness of whether a given session was SHAM vs. tSCS was not formally
190 quantified, and it is possible that participants were aware of that day's intervention. This is a
191 common limitation in tSCS studies, as it is difficult to blind a participant to the sensation vs.
192 absence of electrical stimulation. Note that the first SHAM was always the first session, serving
193 as a wash-in, while the order of the second SHAM was randomized along with the other
194 parameter combinations and used for analysis.

195 In each of these six sessions, participants always underwent a set of spinal motor evoked
196 response testing first, followed by overground gait assessments ("PRE"). They then walked on
197 the treadmill, receiving SHAM or a STIM combination for a total of 25 minutes, including 2.5
198 minutes warm up and 2.5 minutes cool down. After stepping off the treadmill, they rested in a
199 chair for at least five minutes and then repeated the overground gait and RMT assessments
200 ("POST").

201 Overground Gait Assessment

202 To evaluate potential changes in gait, the 10-meter walk test (10MWT) was performed
203 PRE and POST treadmill gait training with inertial measurement units (IMU's) and EMG
204 sensors placed on the participant. The IMU's (XSENS, Enschede, Netherlands) sampled at
205 100Hz and were placed on the posterior aspect of the pelvis at the height of the posterior superior
206 iliac spine, the lateral aspect of the left and right thigh and shank, and on the dorsal aspect of
207 each foot proximal to the metatarsal-phalangeal joints. EMG sensors remained in the same

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placement as during RMT assessment throughout the session. The 10MWT was performed three times each at self-selected (SSV) and fast (FV) velocities on a 14m track, with the middle 10m timed. A GAITRite instrumented walkway was placed in the middle 20 feet (~ 6.1 meters) of the 10MWT (GAITRite, Franklin NJ USA).

Data Analysis

Pre-Processing

Overground data were pre-processed individually for each 10MWT trial. EMG data were pre-processed by first de-meaning the signal, then applying a fourth order bandpass filter with cutoff frequencies 10 – 400Hz. This signal was then rectified and filtered with a second order lowpass filter with 5Hz cutoff frequency. IMU data was pre-processed by applying a fourth order lowpass filter with 6Hz cutoff frequency. All filters were applied to the entire 10MWT trial using MATLAB's `filtfilt` function (MathWorks, Natick MA USA).

Next, the EMG and IMU time series were split by gait cycles, using a gait measurement mat to define heel strike timing. Each gait cycle's time series was time-normalized to 101 points using MATLAB's `resample` function.

Step length (m) and swing duration (sec) were obtained directly from the gait measurement mat output.

Symmetry Calculation

We computed a total of 50 features from this dataset (Supplemental Document 2). Keeping track of which values from each feature derived from paretic vs. nonparetic gait cycles

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allowed us to compute gait symmetry. For each feature, gait symmetry was calculated for the i'th pair of gait cycles using the formula:

$$S_i = \frac{abs(P_i - NP_i)}{0.5 * (P_i + NP_i)} * 100$$

Using this formula, perfect symmetry yields a value of zero, while the worst possible symmetry (assuming positive values for all extracted features) occurs when one side has a value of zero, yielding a symmetry value of 200 [24].

CGAM Calculation

With so many individual symmetry features, we also computed an index of "global" gait symmetry, called the Combined Gait Asymmetry Metric (CGAM). This metric, originally formulated by Ramakrishnan, Lahiff, and Reed [25], is adapted from the Mahalanobis distance, and provides a single aggregate measure of how all of the gait symmetry values together deviate from perfect symmetry and is expressed on the same scale as its constituent symmetry values (0-200). It has previously been used to quantify "global" gait symmetry in one study of gait in persons with stroke [26]. To remove highly collinear variables, we computed the Variance Inflation Factor (VIF) [26], removing variables one at a time until all variables with VIF > 5 were removed. 23 features remained after this procedure: six from the gait measurement mat, 12 from IMU data, and five from EMG data. The six features from the gait measurement mat were stance phase percent duration (% of gait cycle), step duration (sec), step width (m), stride duration (sec), stride length (m), stride velocity (m/s). The 12 IMU features were root mean square (RMS, in deg), sagittal-plane range of motion (deg), max/min joint angle timing (% of gait cycle) for each of the ankle, knee, and hip joint angles. Finally, each of the five muscles' peak activity timing (% of gait cycle) were derived from EMG data. This reduced set of 23 features was then used to

calculate CGAM using the formula below, resulting in one CGAM value per gait cycle. The CGAM calculation is repeated for each unique combination of participant (N=20) and intervention (N=5).

$$CGAM = \sqrt{\frac{(Data) * inv\big(cov(Data)\big) * (Data)'}{\sum inv(cov(Data))}}$$

where *Data* is the matrix of symmetry values. The number of rows in the $Data$ matrix is equal to the number of gait cycles across all 10MWT repetitions in one participant and intervention combination (combining PRE and POST data together), and the number of columns is equal to the number of symmetry variables.

Note that the participants and experimental sessions analyzed in this study are the same as those previously reported by our group [27]. That publication focused on machine-learning prediction of individualized stimulation frequency and intensity, whereas the present analysis evaluates the single-session effects of tSCS using conventional gait symmetry and gait speed outcomes.

Change Scores

We quantified the magnitude of single-session changes in each measure by computing a change score between the means of the PRE and POST values within each intervention and speed (SSV and FV), ensuring that positive change scores indicate an improvement in POST vs. PRE for each metric. For each gait symmetry metric, the change score calculation is PRE - POST. With this formula, because perfect symmetry is zero, improved symmetry (reflected by a lower symmetry value) in POST vs. PRE yields a positive change score. In contrast, for gait

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speed, the change score was computed as POST – PRE, such that improved (faster) gait speed in POST vs. PRE also yields a positive change score.

Statistical Analysis

Step length symmetry and swing duration symmetry were the primary reported outcomes, with CGAM and gait speed reported as secondary outcomes. To evaluate the single-session effects of the interventions, for each outcome a two-way repeated measures ANOVA was performed on the average of the data within each subject, intervention, and speed combination. *Timepoint* (PRE, POST) and *Intervention* (SHAM, $30RMT_{min}$, $50RMT_{min}$, 30TOL, 50TOL) are within subject factors. The primary effect of interest was pairwise comparisons within each intervention level of the *Timepoint x Intervention* interaction effect, to assess whether a specific intervention significantly changed from PRE to POST more than any other. When a significant interaction effect was identified, post hoc pairwise comparisons were conducted. To evaluate whether the order of sessions affected results, we also performed a one-way repeated measures ANOVA with factor *SessionOrder* on the change scores. All p-values are uncorrected.

Data from the first SHAM were excluded from the statistical analysis, as it serves primarily as a wash-in, leaving five analyzed interventions. All values are reported as mean ± standard deviation, and statistical significance was set at $\alpha = 0.05$.

Post Hoc Exploratory Analysis of Person-Specific tSCS Parameters

To further explore potential individualized response patterns to different stimulation patterns and generate hypotheses for future studies, an exploratory post hoc analysis was performed. For each metric, the highest change score in each participant across all interventions was classified the “best” STIM out of the four sessions. The same process was repeated with the

lowest change score to select the "worst" STIM for each of the four sessions. To determine if there was a difference between the best and worst STIM compared to SHAM, a paired t-test was conducted.

Metric	Time-point	SHAM	$30RMT_{min}$	$50RMT_{min}$	30TOL	50TOL
Step Length Symmetry	PRE	35.28 (19.27-51.30)	35.54 (20.47-50.60)	36.91 (20.42-53.39)	30.66 (19.90-41.41)	34.01 (20.34-47.68)
	POST	34.07 (18.43-49.71)	33.95 (18.53-49.37)	31.81 (18.00-45.63)	32.50 (18.73-46.26)	31.33 (18.76-43.90)
Swing Duration Symmetry	PRE	42.37 (33.63-51.11)	42.86 (32.70-53.02)	43.50 (33.64-53.47)	43.33 (34.46-52.19)	44.96 (35.18-54.74)
	POST	41.26 (31.72-50.80)	40.48 (30.19-50.77)	41.42 (31.91-50.92)	42.37 (33.89-51.54)	44.05 (34.35-53.74)
CGAM	PRE	29.78 (22.37-37.2)	29.06 (20.64-37.47)	27.06 (20.36-33.75)	27.99 (21.54, 34.45)	32.52 (23.55- 41.49)
	POST	29.76 (21.41-37.19)	28.94 (20.58, 37.3)	26.92 (20.30, 33.54)	27.82 (21.41, 34.23)	32.40 (23.48, 41.33)
Speed	PRE	0.89 (0.75-1.03)	0.84 (0.69-0.99)	0.84 (0.69-0.99)	0.86 (0.71-1.02)	0.85 (0.71-1.00)
	POST	0.91 (0.76-1.06)	0.89 (0.73-1.05)	0.90 (0.74-1.06)	0.90 (0.74-1.07)	0.90 (0.74-1.05)

Results

Safety

No serious adverse events occurred. A small minority of participants experienced skin redness at the stimulation site which resolved before the next session. Several participants were unable to walk all 25 minutes in one bout, in which case they took either seated or standing rest breaks before resuming. All participants completed all walking sessions within the allotted 40 minutes.

Table 2. Each outcome measure's mean and 95% confidence intervals for each intervention and timepoint combination.

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STIM vs. SHAM

For brevity, results are provided here for FV 10MWT trials only (see Table 2 for mean and 95% confidence interval values). We decided to present FV since it represents the upper limit of participant's motor function, which is likely more sensitive to changes compared to SSV. SSV results are provided in Supplemental Document 3. The two-way repeated-measures ANOVA revealed a significant main effect of *Timepoint* for step length symmetry (avg PRE-POST 1.748 ± 7.372, p=0.0192), swing duration symmetry (avg PRE-POST 1.421 ± 4.423, p=0.0144), CGAM (avg PRE-POST 0.113 ± 0.283, p=0.0038), and 10MWT gait speed (avg POST-PRE 0.043 ± 0.053, p<0.0001) indicating a significant improvement in each outcome measure from PRE to POST-intervention when averaged across all intervention conditions. Neither the main effect of *Intervention* (step length symmetry (p=0.7081); swing duration symmetry (p=0.2713); CGAM (p=0.3044); 10MWT gait speed (p=0.2118)) nor the *Timepoint* × *Intervention* interaction

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(step length symmetry (p=0.0564); swing duration symmetry (p=0.6215); CGAM (p=0.5350); 10MWT gait speed (p=0.4828)) reached statistical significance for any outcome.

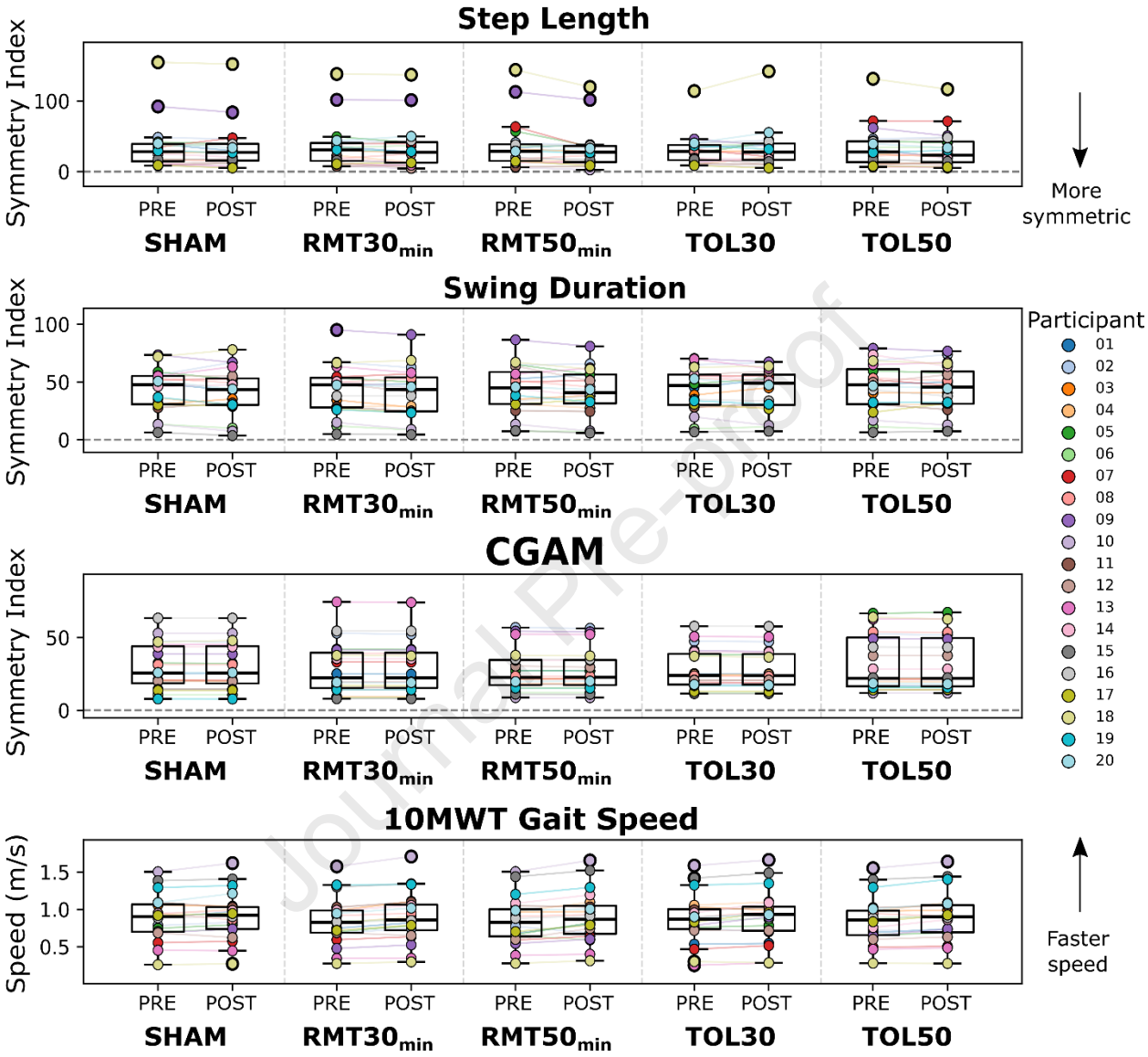


Figure 2. Box plot of mean values PRE and POST for each intervention for each outcome measure. Each point is one participant. Lower symmetry values indicate better symmetry, while higher gait speed values indicate faster gait speed.

Session Order

The order that the interventions occurred did not have a significant effect on the change scores of step length symmetry (p=0.4297), swing duration symmetry (p=0.5357), CGAM (p=0.9804), or 10MWT gait speed (p=0.5303). Notably, auxiliary analyses revealed a

statistically significant increase in baseline 10MWT FV speed across sessions ($p<0.0001$), reflecting the benefits of the treadmill sessions increasing the daily step count of these largely sedentary individuals. This baseline gait speed change was not accompanied by any statistically significant baseline gait symmetry changes. By contrast, for SSV, all metrics besides CGAM ($p=0.1693$) showed significant baseline changes with session order (step length symmetry $p=0.0382$; swing durations symmetry $p=0.0046$; 10MWT speed $p<0.0001$).

Exploratory Analyses

Best Stimulation Parameter vs. SHAM Stimulation

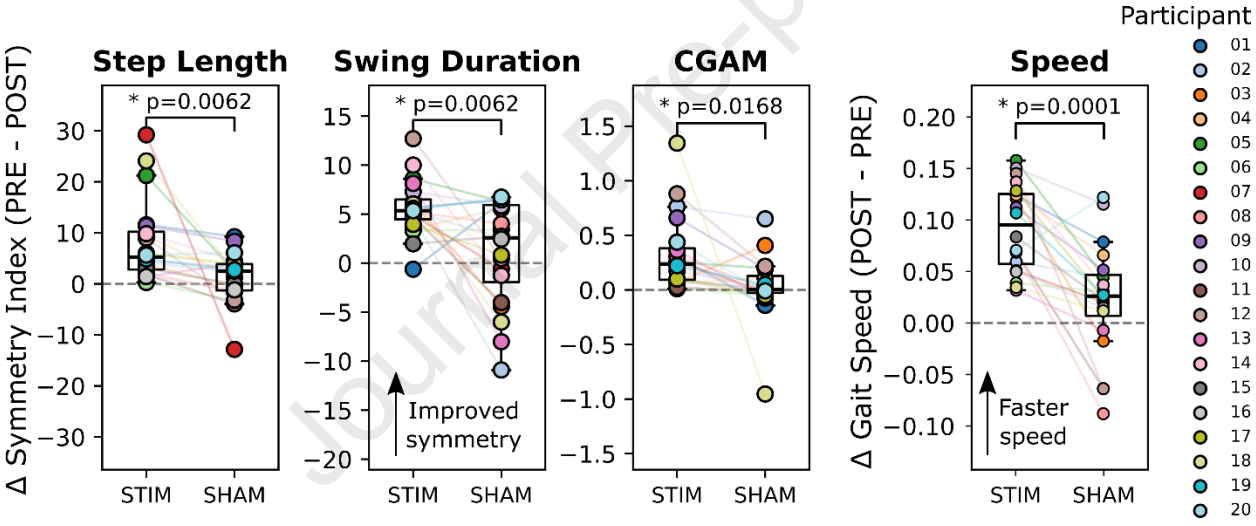
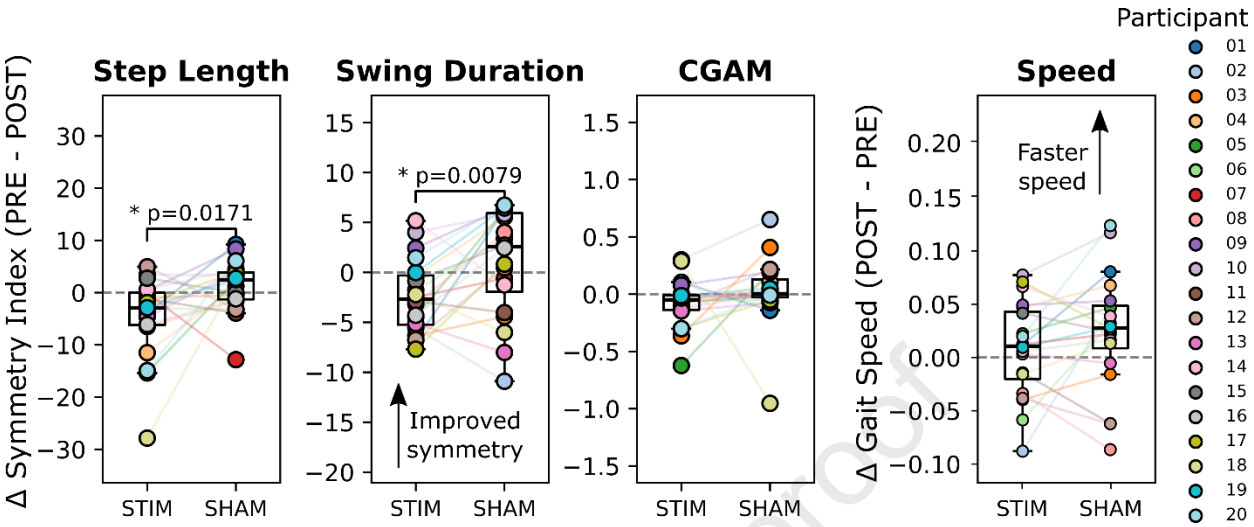


Figure 3. Box plot of PRE vs. POST change scores (mean of POST vs. mean of PRE) of the best STIM and SHAM values for each outcome measure. Each point is one participant. For all measures, positive values indicate improvement: more symmetric, and faster gait speed, respectively.

When comparing the SHAM group with the best of four STIM interventions, a paired t-test found significant improvements in step length symmetry ($\Delta = 7.01 \pm 10.19$, $p=0.0062$), swing duration symmetry ($\Delta = 4.52 \pm 6.56$, $p=0.0062$), CGAM ($\Delta = 0.304 \pm 0.519$, $p=0.0168$), and 10MWT gait speed ($\Delta = 0.069 \pm 0.06$ m/s, $p=0.0001$) (Figure 3).

337 Worst Stimulation Parameter vs. SHAM Stimulation



338

339 *Figure 4. Box plot of PRE vs. POST change scores (mean of POST vs. mean of PRE) of the worst STIM and SHAM values for*
340 *each outcome measure. Each point is one participant. For all measures, positive values indicate improvement: more symmetric,*
341 *and faster gait speed, respectively.*

342 When comparing the SHAM group with the worst STIM intervention, a paired t-test
343 found significant deterioration in step length symmetry (Δ = -5.815 ± 9.956, p=0.0171) and
344 swing duration symmetry (Δ = -3.5 ± 5.27, p=0.0079). There was no significant difference in
345 CGAM (Δ = -0.106 ± 0.406, p=0.2594) or 10MWT gait speed (Δ = -0.018 ± 0.05 m/s, p=0.1330)
346 (Figure 4).

347 # Discussion

348 This study examined the single day effects of transcutaneous spinal cord stimulation
349 (tSCS) plus moderate intensity gait training for people with chronic stroke. We aimed to (1)
350 determine whether tSCS during a single 25-minute bout of treadmill training modulated gait
351 symmetry, and (2) determine whether gait symmetry modulation differed across four

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352 combinations of tSCS frequency (30Hz or 50Hz) and intensity (30% of the minimum of the
353 lower limb muscles' resting motor threshold, or to tolerance). Overall, at the group level, when
354 utilizing the same tSCS parameters for each person, this study found that 25 minutes of tSCS
355 plus treadmill training did not significantly improve step length symmetry, swing duration
356 symmetry, Combined Gait Asymmetry Metric (CGAM), or gait speed during 10MWT compared
357 to treadmill training alone.

358 Changes in motor function are most readily observed after repeated, longitudinal training,
359 especially in people who are several years post-injury or stroke. However, although we did not
360 observe gait symmetry improvements PRE vs. POST STIM relative to SHAM, previous work
361 suggests that tSCS could induce changes in motor function. One study in people < 1 year since a
362 stroke observed changes in gait timing symmetry PRE vs. during tSCS [28]. Another study in
363 people with chronic stroke combined tSCS with constraint force-induced forced use, and found
364 greater increases in paretic step length and height (maximum vertical ankle displacement)
365 compared to the sham condition [29]. In adults with SCI, gait kinematic changes have been
366 observed during tSCS as well [17]. Other studies in children [30] and adults [31] with SCI showed
367 improvements in sitting posture while receiving tSCS. Adults with multiple sclerosis have also
368 shown immediate improvements in postural stability with vs. without tSCS while standing with
369 eyes closed [32]. While these studies suggest that one bout of tSCS can modulate functional
370 outcomes in various clinical populations, they largely investigated the "orthotic" tSCS induced
371 effects (PRE vs. during tSCS). In contrast, the present study investigated "therapeutic" gait
372 symmetry changes PRE vs. POST one 25-minute tSCS session, supporting the goal of tSCS
373 therapy to improve gait even after tSCS is removed. Notably, the only study known to the
374 authors to explore PRE vs. POST changes in one session of tSCS in individuals post-stroke

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reported changes in gait speed of similar magnitude as observed in this study (Skvortsov et al. 0.03 m/s, present study 0.048 m/s), as well as range of motion improvements [33]. Note that the gait speed increases in both this study and Skvortsov et al. were below the "small meaningful change" minimal clinically important difference (MCID) of 0.06 m/s for people with chronic stroke [34].

The non-significant *Timepoint x Intervention* interaction effect for any metric suggests that no set of tSCS parameters influenced any metric's PRE vs. POST change differently than any other set of parameters, similar to previous findings from our group [27]. We interpret this as evidence that for a single-session "therapeutic" effect of tSCS in people with chronic stroke, there may be no universal "best" tSCS parameter. That is not to say that tSCS parameter settings are unimportant – while tSCS parameters showed no differences when the same parameters were grouped across all people, exploratory analysis revealed that the STIM session using the "best" parameters for each person significantly improved gait symmetry relative to SHAM, and similarly the "worst" STIM session exhibited a statistically significant deterioration in gait symmetry relative to SHAM, with the exception of CGAM where this trend did not reach statistical significance. This exploratory analysis leads to a hypothesis that perhaps tSCS-augmented rehabilitation may benefit from person-specific parameters. Indeed, we have previously found that the "best" stimulation parameters may be predicted by baseline participant characteristics [27]. However, these conclusions should be treated as exploratory and hypothesis-generating, as they are confounded by the experiment's design containing four days of STIM sessions but only one SHAM session, leaving open the possibility that these findings are the result of inherent day-to-day variability rather than the effect of tSCS. Future work should explicitly design studies to explore whether person-specific tSCS parameters provide functional

benefit. Other limitations include that (1) we did not assess participants' subjective perceptions of tSCS sensations each day, including whether participants were aware of which session was SHAM, (2) the individuals in our cohort were relatively sedentary and therefore exhibited improvements over time at each session's baseline, although we do not have reason to suspect that this affected this analysis of within-session changes, and (3) the possibility that post-training fatigue may have obscured any beneficial effects. However, within-session 10MWT gait speed increased after training in this study, suggesting that the participants were not overly fatigued.

An additional potential limitation of this study is the focus on spatiotemporal measures (step length, swing duration). Gait spatiotemporal measures are functions of the entire chain of physiological processes that generate movement – signals originate in the brain, travel down the spinal cord, coordinate muscle activities, which in turn modulate movement kinematics, thereby determining foot positioning and timing. Each step in this process is known to be noisy and non-deterministic [35]. Conceivably, measurements taken closer to where movement originates may be more sensitive to tSCS-induced changes. One piece of evidence to support this hypothesis is that as reported above, average CGAM – which includes muscle activity, kinematics, and spatiotemporal measures – significantly improves by 0.11 from PRE to POST training, whereas when it is recomputed using kinematic measures only, it statistically significantly improves by 0.36 ($p=0.0038$). But when CGAM is computed with spatiotemporal measures only, it does not statistically significantly improve from PRE to POST ($p=0.3754$). Note that the magnitude of these changes after one session are still so small as to be likely clinically insignificant, though symmetry does not have an established MCID to reference. Similarly, gait speed, which showed a significant though small increase, is known to be a crucial biomechanical factor, likely reflective of multiple physiological processes as evidenced by its ability to predict mortality [36,37].

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These would suggest that future work on the effects of tSCS on gait symmetry may benefit from focusing on more central changes, as measured by EMG or joint angles, rather than solely spatiotemporal measures.

Finally, this study focused only on the PRE- to POST-tSCS change in gait. Many studies take the alternative route and focus exclusively on the immediate effects of tSCS while it is applied. Both aspects are important, but future research could benefit from more comprehensive studies that include both the immediate and the PRE to POST effects of tSCS on gait.

Conclusion

Our data demonstrate that 25 minutes of combined tSCS and treadmill gait training did not significantly improve step length symmetry, swing duration gait symmetry, the Combined Gait Asymmetry Metric (CGAM), or 10MWT gait speed beyond treadmill gait training alone. We also found no evidence for a universal "best" combination of tSCS burst frequency and intensity to improve gait symmetry. Future work should explore whether longitudinal gait training with tSCS elicits greater gait symmetry changes, and explore the hypothesis that person-specific tSCS parameters may yield larger changes than universally applied parameters. Future studies on tSCS may also benefit from focusing on changes in muscle activity and joint kinematics.

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PUBLIC COMMENTS

This study addresses a clinically relevant and insufficiently characterized question: whether a single session of transcutaneous spinal cord stimulation produces an acute carry-over effect on gait symmetry in people with chronic stroke. Its randomized, sham-controlled, within-participant design and objective gait assessments provide useful, controlled evidence in an area where stimulation protocols and expected clinical effects remain uncertain. Importantly, the absence of a significant primary effect is informative and helps define the limits of single-session tSCS rather than overstating its efficacy. The exploratory parameter analyses are appropriately presented as hypothesis-generating and may help guide future adequately powered studies. Although the modest sample size and acute study design limit broader clinical conclusions, the manuscript provides a scientifically sound and worthwhile contribution to the neuromodulation and stroke rehabilitation literature.

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This study addresses a practical question in stroke rehabilitation: whether a single session of transcutaneous spinal cord stimulation adds measurable benefit to treadmill gait training. In 20 participants with chronic stroke, the intervention did not produce a statistically significant additional improvement in gait symmetry or walking speed compared with training alone.

A clinically relevant feature is that gait was assessed after stimulation had been discontinued. The findings therefore concern the persistence of effects after a single session and should be interpreted within that setting. They do not establish whether stimulation facilitates walking during its application or whether repeated sessions provide benefits over a longer course of

rehabilitation.

Individualized parameter selection remains an interesting direction for further investigation. Its clinical value will depend on demonstrating that individual responses are reproducible and that parameters selected prospectively produce meaningful functional gains. This study helps frame those questions while providing a useful, controlled assessment of the immediate effects of tSCS combined with gait training.

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